Dual-Branch Spectral–Spatial Attention Network for Hyperspectral Image Classification

Jinling Zhao, *Member, IEEE*, Jiajie Wang, Chao Ruan, Yingying Dong, and Linsheng Huang

***Abstract*— In order to achieve accurate hyperspectral image (HSI) classification, the convolutional neural network (CNN) has been extensively utilized. However, most existing patch-based CNN methods overlook the relationship between central pixels and their surroundings. A novel dual-branch spectral–spatial attention network (DBSSAN) is proposed, which helps suppress the impact from interference elements and enhances effective feature extraction from complex features in HSI data. The global and local spatial features are fully integrated through the proposed spatial self-attention module. More specifically, it measures the relationship between the central and surrounding pixels based on cosine similarity and Gaussian–Euclidean similarity to extract global features, while the scale information extraction (SIE) model captures the local features. Furthermore, the inclusion of Transformer model enables the extraction of spectral information from a global perspective, facilitating the capture of long-distance dependencies and nonlinear correlations in HSI. The extracted spectral and spatial features are subsequently classified using a multilayer perceptron (MLP). Five publicly available hyperspectral datasets were used to present experimental evaluations, namely, Indian Pines, Kennedy Space Center, Pavia University, Houston2013, and Houston2018. The comparative results demonstrate the superior performance of the proposed network compared to several state-of-the-art methods.**

***Index Terms*— Central pixels, hyperspectral classification, multilayer perceptron (MLP), spectral–spatial attention, Transformer.**

I. Introduction

HYPERSPECTRAL images (HSIs) contain richer spectral–spatial information compared to ordinary images, making them valuable for capturing detailed material features. Consequently, HSI has been widely applied in fields, such as environmental monitoring, urban land changes, and crop analysis [1], [2], [3]. HSI classification (HSIC) serves as a fundamental requirement for these applications. Early studies in HSIC primarily relied on classic machine learning methods for feature extraction from HSI. These methods included the k-nearest neighbor method [4], the Bayesian estimation method [5], multinomial logistic regression [6], [7], support vector machine (SVM) [8], and random forest (RF) [9]. In addition, various dimensionality reduction and spectral feature extraction techniques were developed in the HSIC field, such as principal component analysis (PCA) [10], independent component analysis (ICA) [11], and others. However, these methods overlooked the spatial correlations among pixels, thus failing to fully exploit spatial features. To address this limitation, various morphological operators were devised, such as morphological profiles and extended multiattribute profiles [12], [13], [14]. Nevertheless, these machine learning methods exhibit certain limitations, including insufficient feature extraction capabilities and poor applicability to different HSIC scenarios.

In recent years, deep learning has emerged as a powerful technique that overcomes the limitations of traditional feature extraction methods. It has the ability to automatically extract deep-level features from data, leading to significant advancements in computer vision tasks, including object detection [15], semantic segmentation [16], and image segmentation [17]. Within the field of HSIC, deep learning models have been extensively explored. Among these models, convolutional neural networks (CNNs) have gained particular attention due to their ability to capture local information and share parameters. CNNs preserve the original data structure while learning features, resulting in a reduced number of network parameters [18]. These advantages have contributed to the superior performance of CNNs in HSIC tasks. For example, Hu et al. [19] proposed a 1-D CNN that effectively extracted spectral features from HSI. Cheng et al. [20] designed an end-to-end 2-D CNN model, utilizing convolutional kernels of varying sizes to capture spatial information. To further enhance feature extraction, Yang et al. [21] combined 2-D and 1-D CNN techniques, constructing a dual-branch network architecture to jointly extract spectral and spatial features. Chen et al. [22] introduced a 3-D CNN that efficiently captures both spectral and spatial characteristics.

While 3-D CNNs are effective in extracting spectral and spatial information from HSI, the high dimensionality and complexity of HSI often require additional layers and parameters to capture features in each dimension. However, this

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can pose challenges in model training, increase the number of parameters significantly, and potentially lead to overfitting. To address these issues, Zhong et al. [23] proposed a spectral–spatial residual network (SSRN) that utilizes residual connections to enhance feature utilization by supplementing the feature information of the front layer with that of the back layer. Paoletti et al. [24] introduced the pyramid residual network for HSI classification, which employs a pyramid-like structure of residual modules to progressively expand feature map dimensions and effectively extract feature information. Wang et al. [25] proposed a fast and dense spectral–spatial convolution network (FDSSC) that addresses overfitting by concatenating the outputs of preceding convolutional layers as the input for each subsequent convolutional layer. Xie et al. [26] developed a multiscale densely connected convolutional network (MS-DenseNet) that incorporates multiple dense blocks to fuse relevant information from different-scale images. To address the challenge of large-scale parameters resulting from 3-D CNN, a two-branch architecture has increasingly become the preferred approach for HSIC. Guo et al. [27] proposed a dual-view spectral and global spatial feature fusion network (DSGSF) aimed at extracting spatial–spectral features. The network integrates the encoder–decoder architecture of channel and spatial attention to learn global spatial features. In addition, the spectral information is divided into large- and small-scale dual-view information, with the use of LSTM units to extract the spectral features. Graph convolutional network (GCN), a variant of CNN, is capable of extracting various types of spectral–spatial features [28]. The attention multihop graph and multiscale convolutional fusion network (AMGCFN) [29] utilizes two branches, comprising multiscale CNN and multihop GCN, to extract multiscale spectral–spatial features. In order to enhance the model's discriminative power, Duan et al. [30] introduced a structure-preserved hyper GCN (SPHGCN). This model consists of branches based on regular convolution and hypergraph convolution, surpassing the limitations of image patches. By aggregating information between multiple HSI patches, it further acquires deep semantic features.

While these methods, based on CNN backbone networks and their variations, have significantly improved the performance of HSIC, it is important to note that CNN's convolutional operations treat input content uniformly, despite the varying contributions of spectral and spatial features to HSIC [31]. Furthermore, CNN's focus on extracting features from local neighborhoods limits its ability to capture long-range dependencies between pixels. This limitation becomes apparent when CNN processes HSIs with intricate spatial layouts and structures. In the realm of computer vision, researchers have recognized the benefits of integrating attention mechanisms, which allow for selective focus on salient regions rather than treating all parts equally. By combining CNN with attention mechanisms, the model can concentrate on specific local regions of input data, assigning weights that emphasize important features in the feature maps while attenuating less relevant ones. This integration effectively improves the model's performance by highlighting meaningful information and suppressing less useful features. To enhance spectral information, Cui et al. [32] introduced the squeeze-and-excitation (SE) module [33]. This module employs global pooling to compress and dynamically adjust features across spectral bands, optimizing feature representation. Similarly, Woo et al. [34] proposed the convolutional block attention module (CBAM), which sequentially generates attention maps in both spectral and spatial dimensions. These attention maps are then multiplied with the input feature maps, enabling adaptive feature refinement. To extract more distinctive spectral and spatial features, Ma et al. [35] enhanced CBAM by employing a dual-branch strategy to capture both types of features. Zhu et al. [18] proposed the residual spectral–spatial attention network (RSSAN), which incorporates CBAM within residual blocks. This approach not only extracts spectral and spatial features but also effectively mitigates overfitting issues. In addition, Zhang et al. [36] developed a cross-data-source cross-attention mechanism that effectively integrates multiple data sources, leading to a substantial enhancement in classification accuracy.

When exploring further methods for feature optimization in HSIC, two approaches that have gained significant attention are the Transformer [37] and nonlocal neural networks [38]. The Transformer utilizes self-attention mechanisms to capture relationships between different positions within input sequences. To effectively extract features from HSIs, Sun et al. [39] proposed the spectral–spatial feature tokenization transformer (SSFTT) model, which combines the Transformer and CNN. On the other hand, nonlocal neural networks model long-range dependencies between pixels by computing the global similarity at each position within the image. Sun et al. [40] introduced the 3-D CNN-based spectral–spatial attention network (SSAN), which employs nonlocal neural networks to extract joint spectral–spatial features. While the nonlocal neural network considers the correlation between all pixels, it is important to note that the input of the CNN-based HSI classification model is a patch cube sample, where the initial information of each sample is only the category of the central pixel. Therefore, the input patch is used to predict its central pixel, and neighboring pixels may have different effects on the prediction classification of the central pixel. Hence, it becomes imperative to investigate the potential correlations between the central pixel and its neighboring pixels from a global perspective [31]. To address this, Liu et al. [28] proposed a multiregion target attention model based on the Transformer's multihead attention mechanism. This model is designed to extract both spectral and spatial information from the central pixel and its neighboring pixels within multiscale regions. Liu et al. [41] developed a scaled dot-product central attention and multihead central attention mechanism. This approach extracts spectral and spatial information from the central pixel and its surrounding pixels that are similar to the central pixel. Zhang et al. [42] introduced an attention-based lightweight network (ALN), which employs dense connection blocks to extract spectral features and incorporates a spatial self-attention module to explore the cosine similarity between the central pixel and its neighboring pixels, thus delving into spatial characteristics. In addition, Zhang et al. [31] introduced the spectral–spatial

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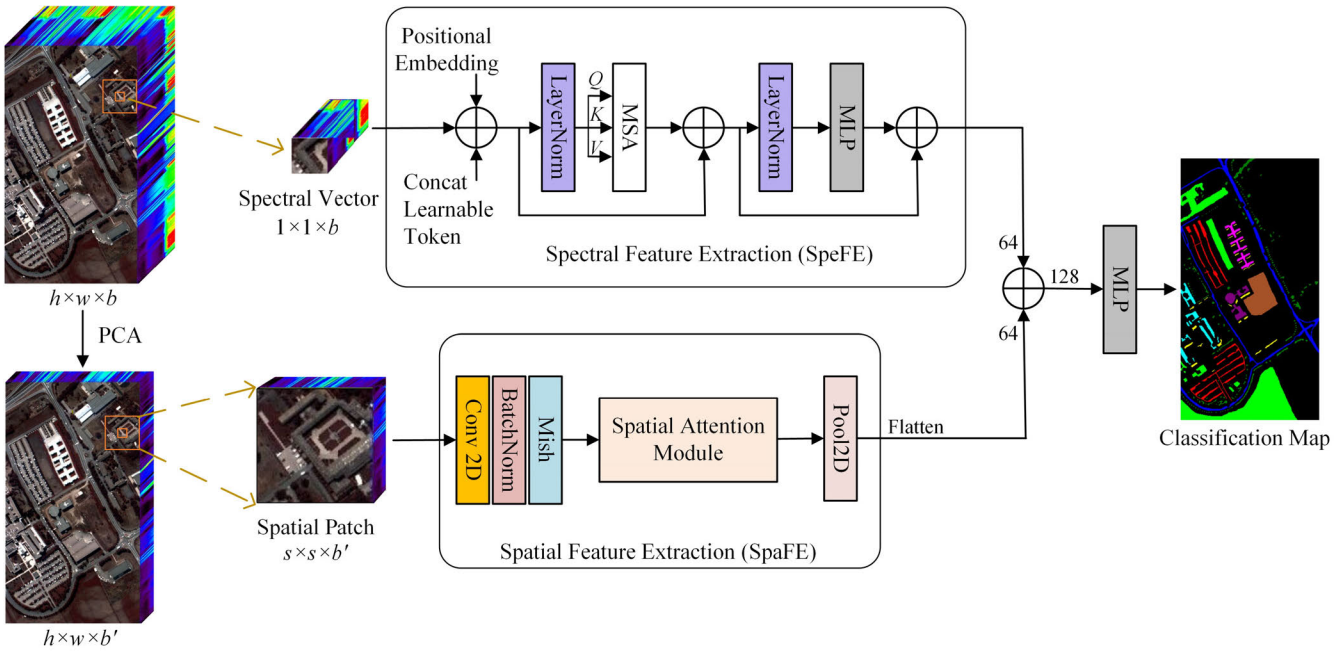


Fig. 1. Overall framework of DBSSAN.

self-attention network (SSSAN), which extracts spatial features by exploring the similarity between the central pixel and its neighboring pixels and extracts spectral features based on the concept of nonlocal neural networks.

While these methods have shown promising results, they all utilize cosine similarity to compute pixel-wise similarity, which only considers the direction of the feature vector and not its length. In other words, as long as two vectors are in the same direction, regardless of their length or degree, they are considered similar. To address this concern, Li et al. [43] combined cosine similarity and Euclidean similarity, successfully mitigating the insensitivity of cosine similarity to the absolute magnitudes of features. However, it is important to note that Euclidean similarity faces the curse of dimensionality in high-dimensional data, where its effectiveness diminishes as the data dimension increases. This can result in an inability to capture the intrinsic structure and feature variations of high-dimensional data. Therefore, in HSIC tasks, Euclidean similarity might not adequately differentiate between data points of different classes, leading to a decrease in classification performance.

To address the limitations mentioned above, this study proposes a dual-branch spectral–spatial attention network (DBSSAN) for HSI classification. The DBSSAN consists of two branches that focus on extracting spatial and spectral features. In the spatial branch, the spatial attention module is employed to extract spatial features. This module incorporates the cosine and Gaussian–Euclidean similarity self-attention module (CG-SA) to capture global features and the scale information extraction (SIE) module to capture local features. By integrating global and local features, the DBSSAN enhances the expression of spatial characteristics. In the spectral branch, the spectral information is transformed into a 1-D sequence and fed into the Transformer module to extract spectral features. The Transformer module effectively captures the relationships among different spectral components and avoids the need for 3-D CNN-based feature extraction. This approach reduces the model's parameter count and computational complexity while mitigating the risk of overfitting. Finally, the spatial and spectral features extracted from the two branches are fused using a multilayer perceptron (MLP) for classification. The main contributions of this study can be summarized as follows.

1) The introduction of the CG-SA module enables a comprehensive exploration of the similarity between the central pixel and its neighboring pixels while mitigating the curse of dimensionality. By mapping the HSI into a higher dimensional space, the CG-SA module enhances spatial feature extraction.
2) The incorporation of the SIE module allows for the fusion of global and local information, enabling the model to acquire rich contextual information and enhance its feature extraction capability.
3) The design of the Transformer module for spectral feature extraction not only effectively captures relationships among different spectral components but also reduces the model's parameter count and computational complexity while decreasing the risk of overfitting.

II. Methodology

A. Overview

The proposed network's overall framework is illustrated in Fig. 1. We denote the HSI dataset as $\boldsymbol{H} \in \mathbb{R}^{h\times w\times b}$, where h, w, and b represent the length, width, and number of bands of the dataset, respectively. To ensure computational efficiency and reduce costs, we perform dimensionality reduction on $\boldsymbol{H}$ using PCA, resulting in $\boldsymbol{H}_{\text{PCA}} \in \mathbb{R}^{h\times w\times b'}$, where b' represents the

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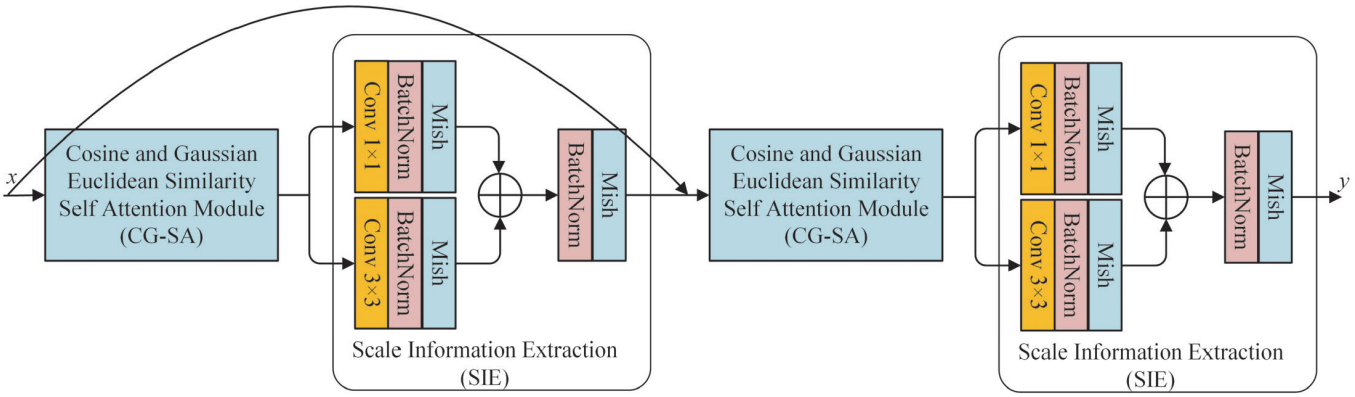


Fig. 2. Spatial attention module.

number of bands after dimensionality reduction. Subsequently, $\boldsymbol{H}_{\mathrm{PCA}}$ is divided into cubic blocks, which serve as the input for spatial feature extraction. Initially, a 2-D convolutional layer is applied to capture sufficient input information. Then, the spatial attention module is utilized to fuse global and local features. Finally, a global 2-D average pooling operation is performed to obtain 1-D spatial features. Simultaneously, the spectral input of is processed by the Transformer module to obtain 1-D spectral features. Ultimately, the extracted spatial and spectral features are inputted into an MLP for classification. The following sections will provide a detailed explanation of the spatial and spectral feature extraction processes.

B. Spatial Feature Extraction

Global feature extraction is crucial for capturing the contextual information of the entire HSI, enabling the model to understand its overall structure and distribution. On the other hand, local feature extraction focuses on capturing fine details and subtle variations, which is essential for distinguishing objects with similar spectra. To enhance feature representation, this study proposes a spatial attention module that combines global and local features. Fig. 2 illustrates the proposed spatial attention module, which consists of two components: the cosine and Gaussian–Euclidean similarity self-attention (CG-SA) module for global feature extraction and the SIE module for local feature extraction. The CG-SA module explores the relationship between the central pixel and its neighboring pixels to extract global features. Meanwhile, the SIE module employs convolution kernels of varying sizes to extract local features. To mitigate the risk of overfitting, a residual connection is established between the input data and the fused data of global and local information, followed by feature fusion.

1) The CG-SA module specifically investigates the relationship between the central pixel and its neighboring pixels using both cosine similarity and Gaussian–Euclidean similarity. Cosine similarity measures the angle between the central pixel and its neighboring pixels to evaluate their similarity in feature space, potentially defining them as similar when their directions align. Meanwhile, Gaussian–Euclidean similarity retains the benefits of Euclidean similarity by accounting for both the magnitude and direction of vectors. In addition, the introduction of the Gaussian kernel helps alleviate the dimensionality curse associated with Euclidean similarity. This article integrates both cosine similarity and Gaussian–Euclidean similarity to comprehensively examine the relationship between the central pixel and its neighboring pixels. As depicted in Fig. 3, the input to the module is denoted as $\boldsymbol{X} \in \mathbb{R}^{s\times s\times c}$, where $s \times s$ represents the spatial size and c indicates the number of channels. The center pixel of $\boldsymbol{X}$ is represented as $\boldsymbol{X}_i \in \mathbb{R}^{1\times 1\times c}$, and the surrounding pixels are denoted as $\boldsymbol{X}_{i,t} = [\boldsymbol{X}_{i,1}, \boldsymbol{X}_{i,2}, \boldsymbol{X}_{i,3}, \ldots, \boldsymbol{X}_{i,n}]$ with $n = s \times s$. The Gaussian–Euclidean similarity and cosine similarity between the central pixel $\boldsymbol{X}_i$ and surrounding pixels are denoted as $\boldsymbol{G}_{i,t}$ and $\boldsymbol{C}_{i,t}$

$$\begin{aligned}\boldsymbol{G}_{i,t} &= \mathrm{GESim}(\boldsymbol{X}_i, \boldsymbol{X}_{i,t}) \\ &= \exp\left(-\frac{\left\|\boldsymbol{X}i - \boldsymbol{X}_{i,t}\right\|^2}{2\sigma^2}\right), \quad t = 1, 2, \ldots, n \quad (1)\end{aligned}$$

where σ affects the rate of similarity decay, in which the initial sigma of this model is 1, and adaptively adjusts when the model is optimized

$$\begin{aligned}\boldsymbol{C}_{i,t} &= \mathrm{Cossim}(\boldsymbol{X}_i, \boldsymbol{X}_{i,t}) \\ &= \left(\frac{\boldsymbol{X}_i \boldsymbol{X}_{i,t}^T}{\|\boldsymbol{X}_i\|\left\|\boldsymbol{X}_{i,t}\right\|}\right)^2, \quad t = 1, 2, \ldots, n. \quad (2)\end{aligned}$$

When the values of $\boldsymbol{G}_{i,t}$ and $\boldsymbol{C}_{i,t}$ approach 1, it means that they are more similar; on the contrary, when the value approaches 0, the similarity gradually decreases. Further normalization using the softmax function, from the corresponding similarity matrix, the Gaussian–Euclide distance self-similar attention map $\boldsymbol{GaEd} \in \mathbb{R}^{s\times s}$ and the cosine self-similar attention m map $\boldsymbol{Cos} \in \mathbb{R}^{s\times s}$ are obtained, respectively,

$$\begin{aligned}\boldsymbol{GaEd} &= \mathrm{softmax}(\boldsymbol{G}_{i,t}) \\ &= \frac{\exp(\boldsymbol{G}_{i,t})}{\sum_{t=1}^{n}\exp(\boldsymbol{G}_{i,t})}, \quad t = 1, 2, \ldots, n \quad (3) \\ \boldsymbol{Cos} &= \mathrm{softmax}(\boldsymbol{C}_{i,t}) \\ &= \frac{\exp(\boldsymbol{C}_{i,t})}{\sum_{t=1}^{n}\exp(\boldsymbol{C}_{i,t})}, \quad t = 1, 2, \ldots, n. \quad (4)\end{aligned}$$

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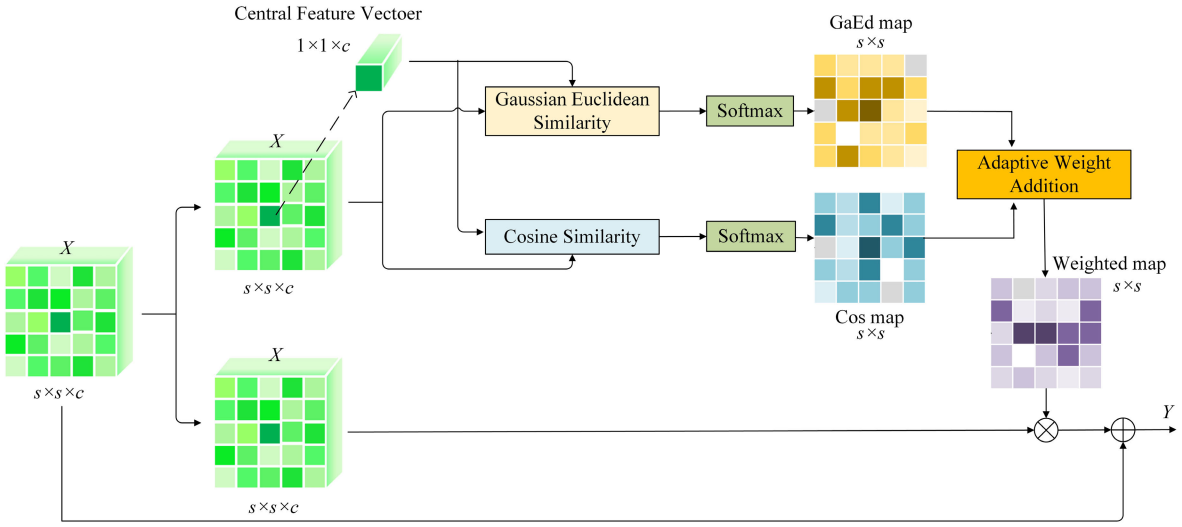


Fig. 3. Structure of the proposed cosine and Gaussian–Euclidean similarity self-attention (CG-SA) module.

In addition, an adaptive weight addition pattern is used to fuse two similar attentions, thereby enhancing the representation of spatial information, as detailed in the following:

$$\boldsymbol{Weighted} = \lambda \times \boldsymbol{GaEd} + (1 - \lambda) \times \boldsymbol{Cos} \tag{5}$$

where $\boldsymbol{Weighted} \in \mathbb{R}^{s \times s}$ is fusion similarity attention. λ is a weighted parameter with an initial value of 0.5 that can be optimized adaptively when the model is optimized.

Finally, the fused similarity attention is element-wise multiplied with the input $\boldsymbol{X}$ along the spectral channel dimension, resulting in relevant information features associated with the central pixel while suppressing unnecessary features. This is followed by a residual connection to prevent gradient vanishing and network degradation, as detailed in the following:

$$\boldsymbol{Y} = \boldsymbol{X} \otimes \boldsymbol{Weighted} + \boldsymbol{X}. \tag{6}$$

2) *SIE Module:* As depicted in Fig. 2, the SIE module is designed to further extract spatial local features using a dual-branch approach, employing 1 × 1 and 3 × 3 convolutions. Specifically, the scale branch includes a 1 × 1 convolution layer, a batch normalization (BN) layer, and a Mish activation function. In addition, the 3 × 3 convolution branch consists of a 3 × 3 convolution layer, a BN layer, and a Mish activation function. The features from these two scale branches are then fused using an element-by-element addition operation, enabling comprehensive extraction and complementarity of scale information. The inclusion of the Mish activation function ensures a certain level of smoothness while introducing nonlinearity, effectively mitigating the issue of disappearing gradients, and improving training stability.

C. Spectral Feature Extraction

This manuscript presents a novel approach for HSIC, employing a Transformer module to extract spectral information, as depicted in Fig. 1. In this article, the use of Transformer in conjunction with CNN deviates from traditional methods. The Transformer network completely eliminates convolutional operations and, in their place, utilizes a pure attention mechanism to capture global dependencies within input sequences, showcasing significant scalability. Furthermore, it circumvents the need for redundant convolutional operations, resulting in reduced computational costs. In addition, the Transformer module offers several advantages for extracting spectral information from HSI data.

First, the self-attention mechanism of the Transformer allows the model to establish intricate relationships between spectral bands on a global scale, facilitating the capture of comprehensive contextual information within the data. This global context is crucial for accurate classification.

Second, the adaptive weight learning capability of the Transformer enables selective focus on important spectral features, enhancing the discriminative power of the model. This adaptability ensures that the model prioritizes relevant information while suppressing noise and irrelevant features. Furthermore, the self-attention mechanism of the Transformer effectively handles long-range dependencies within the data. By capturing associations between distant pixels, the model can better understand the spatial relationships and patterns present in the HSI.

Overall, the proposed Transformer module offers a promising solution for HSIC, providing efficient spectral feature extraction and capturing both local and global context information.

In recent years, the self-attention mechanism (SA) has been modeled after the human retina, focusing on more crucial information. Furthermore, methods based on attention mechanisms can capture long-range dependencies, enabling the extraction of global features, which has found extensive

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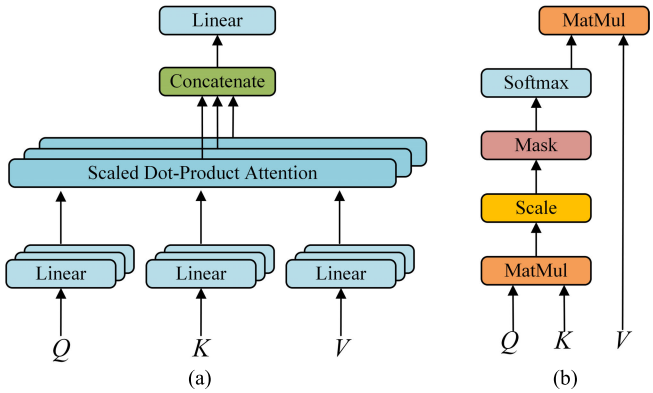


Fig. 4. Illustration of the attention mechanism in the Transformer. (a) MSA. (b) SA.

applications in HSI classification. The structure of SA is illustrated in Fig. 4(b). As seen, SA involves three inputs, namely, $\boldsymbol{Q}$ (query), $\boldsymbol{K}$ (key), and $\boldsymbol{V}$ (value). Subsequently, the correlation between $\boldsymbol{Q}$ and $\boldsymbol{K}$ is multiplied to obtain the relevance at different positions. Finally, the obtained relevance matrix is multiplied with $\boldsymbol{V}$. This mechanism allows for the weighting of the importance of information, generating feature outputs with different attention levels. In other words, SA strengthens essential features while suppressing irrelevant ones.

To mitigate the potential drawback of the SA excessively focusing on its own position when encoding current location information in the model, a multihead self-attention mechanism (MSA) has been introduced. MSA, as depicted in Fig. 4(a), connects multiple SA modules, allowing the same attention mechanism to learn different behaviors by utilizing different subspaces representing queries, keys, and values. This approach actively learns to focus on diverse content

$$\mathrm{SA} = \mathrm{Attention}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \mathrm{softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}}{\sqrt{dk}}\right)\boldsymbol{V} \tag{7}$$

$$\mathrm{MSA}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \mathrm{Concat}(\mathrm{SA}_1, \mathrm{SA}_2, \mathrm{SA}_3, \ldots, \mathrm{SA}_i)\boldsymbol{W}. \tag{8}$$

Let the spectral information be denoted as $\boldsymbol{H}_{\mathrm{spe}} \in \mathbb{R}^{1\times 1\times b}$. Simultaneously, it is connected to a learnable classification token $\boldsymbol{T}^{\mathrm{cls}}$. In addition, to differentiate the positions of each spectral band, a position encoding $\boldsymbol{PE}_{\mathrm{pos}}$ is introduced, resulting in the final input sequence

$$\boldsymbol{X}_{\mathrm{spe}} = \left[\boldsymbol{T}^{\mathrm{cls}}, \boldsymbol{H}_{\mathrm{spe1}}, \boldsymbol{H}_{\mathrm{spe2}}, \boldsymbol{H}_{\mathrm{spe3}}, \ldots, \boldsymbol{H}_{\mathrm{spe}b}\right] + \boldsymbol{PE}_{\mathrm{pos}}. \tag{9}$$

$\boldsymbol{X}_{\mathrm{spe}}$ is passed into the subsequent model. The MSA module, as mentioned above, consists of multiple SA components.

In SA, the input sequence $\boldsymbol{X}_{\mathrm{spe}}$ is linearly mapped to three matrices $\boldsymbol{Q}$, $\boldsymbol{K}$, and $\boldsymbol{V}$. Calculate the similarity between $\boldsymbol{Q}$ and $\boldsymbol{K}$ to obtain attention weights and apply these weights to $\boldsymbol{V}$ to calculate context-aware features for each position. The formula is shown in (7) and (8), where dk is the dimension of $\boldsymbol{K}$, i is the number of SA layers stacked, and $\boldsymbol{W}$ is a weight matrix, a learnable parameter that enables the model to adapt to different input data and tasks by learning the values of the weight matrix.

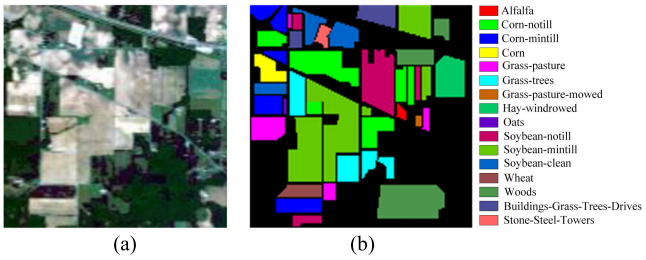


Fig. 5. IP. (a) False-color map. (b) Ground truth map.

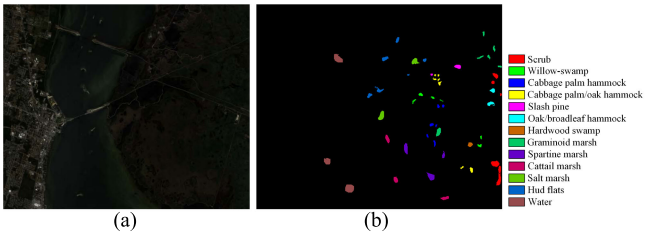


Fig. 6. KSC. (a) False-color map. (b) Ground truth map.

Next, the learned weight matrix obtained from the previous step is fed into the LN and MLP layers. LN aids in reducing internal covariate shifts, while MLP performs additional nonlinear transformations and feature combinations. This process results in the generation of a richer and higher dimensional feature representation.

III. Experiments

A. Datasets Description and Evaluation Metrics

To validate the effectiveness and reliability of the proposed algorithm, five publicly available hyperspectral datasets are selected for experimentation. These datasets include the Indian Pines (IP), Kennedy Space Center (KSC), Pavia University (PU), Houston2013 (HU13), and Houston2018 (HU18) datasets.

The IP dataset was collected in 1992 using the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) at the Indian Pine Agricultural Test Ground in northwestern Indiana, USA. It consists of 220 continuous spectral bands within a wavelength range of 0.4–0.45 μm. The dataset has a size of 145 $\times$ 145 with a spatial resolution of 20 m. It contains a total of 10 249 samples, encompassing 16 different land cover categories. For analysis, 200 bands were used after excluding water absorption and low signal-to-noise ratio bands. Fig. 5(a) illustrates its false-color image, while Fig. 5(b) presents the corresponding ground truth map.

The KSC dataset, collected in 1996 at the Kennedy Space Center in Florida, USA, also utilizes AVIRIS. It encompasses 224 continuous spectral bands within a wavelength range of 0.4–0.45 μm. With a size of 512 $\times$ 614 and a spatial resolution of 18 m, it contains a total of 5211 samples representing 13 different land cover categories. To mitigate noise interference, 176 bands were selected for analysis. Fig. 6(a) displays the false-color image of the KSC dataset, while Fig. 6(b) depicts the associated ground truth map.

The PU dataset, obtained in 2003, represents a campus map of the University of Pavia in northern Italy. It was

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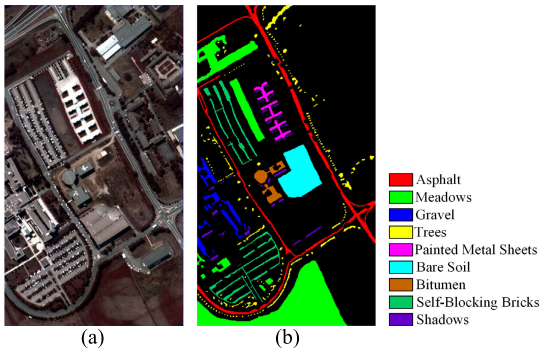


Fig. 7. PU. (a) False-color map. (b) Ground truth map.

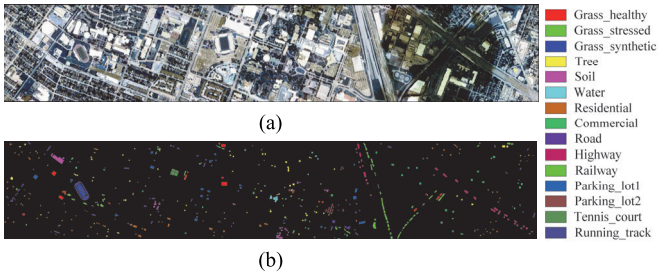


Fig. 8. HU13. (a) False-color map. (b) Ground truth map.

collected using the Reflective Optics Spectrographic Imaging System (ROSIS-03) and comprises 115 continuous spectral bands within a wavelength range of 0.43–0.86 μm. The dataset has a size of 610 $\times$ 340 with a spatial resolution of 1.3 m. It contains a total of 42 776 samples, covering nine different land cover categories. For analysis, 103 bands were selected after eliminating bands susceptible to noise interference. Fig. 7(a) showcases the false-color image of the PU dataset, while Fig. 7(b) provides the corresponding ground truth map.

The HU13 dataset was acquired using the CASI-1500 sensor in 2012 for the 2013 GRSS Data Fusion Contest, covering the University of Houston and its surrounding areas. This dataset comprises 144 spectral bands, ranging from 0.38 to 1.05 μm in wavelength and has dimensions of 349 $\times$ 1905. It includes classifications for five natural and ten artificial object types. Fig. 8(a) displays the false-color image of the HU13 dataset, while Fig. 8(b) shows the corresponding ground truth map.

The HU18 dataset was acquired at the University of Houston in 2017 using the CASI-1500 sensor for the 2018 GRSS Data Fusion Contest. It encompasses 48 spectral bands spanning a wavelength range from 0.38 to 1.05 μm. With dimensions of 601 $\times$ 2384, the dataset encompasses details for 20 distinct land cover categories. Fig. 9(a) displays the false-color image of the HU13 dataset, while Fig. 9(b) depicts the associated ground truth map.

Due to the varying sizes of available labeled samples in the five datasets, different proportions of samples were used for training, validation, and testing. Specifically, for the IP, KSC, and HU13 datasets, we employed proportions of 10%, 1%, and 89% for training, validation, and testing, respectively. As for the PU dataset, proportions of approximately 5%, 0.5%, and 94.5% were used for the same purposes. In addition, the

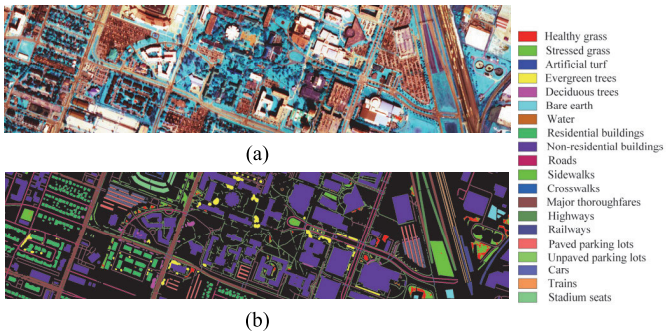


Fig. 9. HU18. (a) False-color map. (b) Ground truth map.

TABLE I
NUMBER OF SAMPLES FOR EACH CLASS IN THE IP DATASET

No.	Class name	Train (10%)	Val (1%)	Test (89%)
1	Alfalfa	4	1	41
2	Corn-notill	142	14	1272
3	Corn-mintill	83	8	739
4	Corn	23	2	212
5	Grass-pasture	48	4	431
6	Grass-trees	73	7	650
7	Grass-pasture-mowed	3	1	24
8	Hay-windrowed	47	4	427
9	Oats	3	1	16
10	Soybean-notill	97	9	866
11	Soybean-mintill	245	24	2186
12	Soybean-clean	59	5	529
13	Wheat	20	2	183
14	Woods	126	12	1127
15	Buildings-Grass-Trees-Drives	38	3	345
16	Stone-Steel-Towers	9	1	83
	Total	1020	98	9131

TABLE II
NUMBER OF SAMPLES FOR EACH CLASS IN THE KSC DATASET

No.	Class name	Train (10%)	Val (1%)	Test (89%)
1	Scrub	76	7	678
2	Willow swamp	24	2	217
3	Cabbage palm hammock	25	2	229
4	Cabbage palm/oak hammock	25	2	225
5	Slash pine	16	1	144
6	Oak/broadleaf hammock	22	2	205
7	Hardwood swamp	10	1	94
8	Graminoid marsh	43	4	384
9	Sprtine marsh	52	5	463
10	Cattail marsh	40	4	360
11	Salt marsh	41	4	374
12	Hud flats	50	5	448
13	Water	92	9	826
	Total	516	48	4647

HU18 dataset was divided into training, validation, and testing sets with proportions of 2.0%, 0.5%, and 97.5%, respectively. Please refer to Tables I–V for specific details.

In the upcoming experiments, we will employ three metrics to assess the classification performance: overall accuracy (OA), average accuracy (AA), and Kappa coefficients (K). OA represents the ratio of correctly classified labeled samples to the total number of samples in the test set, providing an indication of the overall accuracy of the classification. AA measures the accuracy of each class by calculating the

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TABLE III
NUMBER OF SAMPLES FOR EACH CLASS IN THE PU DATASET

No.	Class name	Train (5%)	Val (0.5%)	Test (94.5%)
1	Asphalt	331	33	6267
2	Meadows	932	93	17624
3	Gravel	104	10	1985
4	Trees	153	15	2896
5	Painted Metal Sheets	67	6	1272
6	Bare Soil	251	25	4753
7	Bitumen	66	6	1258
8	Self-Blocking Bricks	184	18	3480
9	Shadows	47	4	896
	Total	2135	210	40431

TABLE IV
NUMBER OF SAMPLES FOR EACH CLASS IN THE HU13 DATASET

No.	Class name	Train (10%)	Val (1%)	Test (89%)
1	Grass_healthy	125	12	1114
2	Grass_stressed	125	12	1117
3	Grass_synthetic	69	6	622
4	Tree	124	12	1108
5	Soil	124	12	1106
6	Water	32	3	290
7	Residential	126	12	1130
8	Commercial	124	12	1108
9	Road	125	12	1115
10	Highway	122	12	1093
11	Railway	123	12	1100
12	Parking_lot1	123	12	1098
13	Parking_lot2	46	4	419
14	Tennis_court	42	4	382
15	Running_track	66	6	588
	Total	1496	143	13390

TABLE V
NUMBER OF SAMPLES FOR EACH CLASS IN THE HU18 DATASET

No.	Class name	Train (2%)	Val (0.5%)	Test (97.5%)
1	Healthy grass	195	48	9556
2	Stressed grass	650	162	31690
3	Artificial turf	13	3	668
4	Evergreen trees	271	67	13257
5	Deciduous trees	100	25	4896
6	Bare earth	90	22	4404
7	Water	5	1	260
8	Residential buildings	795	198	38779
9	Non-residential buildings	4475	1118	218159
10	Roads	917	229	44720
11	Sidewalks	680	170	33179
12	Crosswalks	30	7	1481
13	Major thoroughfares	926	231	45191
14	Highways	197	49	9619
15	Railways	138	34	6765
16	Paved parking lots	230	57	11213
17	Unpaved parking lots	3	1	142
18	Cars	130	32	6385
19	Trains	107	26	5236
20	Stadium seats	136	34	6654
	Total	10088	2514	492254

ratio of classification accuracy for each category to the total number of categories in the dataset. It serves as a measure of the AA across all classification classes. The Kappa coefficient offers a comprehensive evaluation of the consistency between the classification result and the ground truth. Higher values for these metrics indicate better classification performance. To analyze the efficiency of the algorithms quantitatively, we evaluate the running time of each algorithm by measuring the training time (Ttrain) and test time (Ttest). In addition, the model complexity and computational cost are assessed separately based on the size of model parameters (Params) and floating-point operations (FLOPs).

B. Experimental Configuration

For consistency in our experiments, we conducted all tests on laptops equipped with an AMD R7-5800 processor, 16 GB of RAM, and an NVIDIA GeForce RTX 3060 laptop GPU with 6 GB of video memory. The software environment consisted of a 64-bit Windows 11 operating system, Python 3.7, and the PyTorch 1.12 deep learning framework.

C. Parameter Setting

Before proceeding with the experiments, we applied standardized preprocessing to all datasets [43]. This involved normalizing the data to have a mean of 0 and a variance of 1. By doing so, we eliminated dimensional differences between different features, enhancing the model's stability and generalization ability to input data. Furthermore, this preprocessing step reduced sensitivity to noise changes, making the data easier to process and compare using the algorithm. Subsequently, we performed PCA dimensionality reduction on the preprocessed data. For the IP, KSC, PU, HU13, and HU18 datasets, we reduced the dimensions to 30, 18, 4, 4, and 5, respectively. This step helped reduce the data dimensions, extract the primary features, and further simplify the model's complexity.

In the proposed training network, we set the batch size to 64 and the number of training iterations to 150. We utilized the Adam optimizer [44] to update the gradient and employed cross-entropy loss as the loss function for network optimization. In addition, we conducted a detailed analysis of key hyperparameters to further optimize the classification performance. The following section provides a comprehensive overview of the experiment details.

1) Learning Rate: It plays a critical role in the training process as it determines the convergence speed and effectiveness of the model. If the learning rate is too large, it may lead to missing the optimal solution and causing oscillations. Conversely, if the learning rate is too small, the training convergence speed is slow. To identify the optimal learning rate for each dataset, this experiment explores learning rates of $\{0.01, 0.003, 0.001, 0.0003, 0.0001\}$. The results are presented in Fig. 10(a)–(c). The optimal learning rate for the IP, PU, and HU13 datasets was found to be 0.003, whereas the KSC and HU18 datasets achieved peak performance at 0.001. Consequently, for subsequent experiments, the learning rate was set to 0.003 for IP, PU, and HU13, and 0.001 for KSC and HU18.

2) Spatial Input Size: It determines the amount of spatial information surrounding a pixel used for classification. A larger spatial window includes more spatial neighbor information, which can capture more complex spatial patterns and

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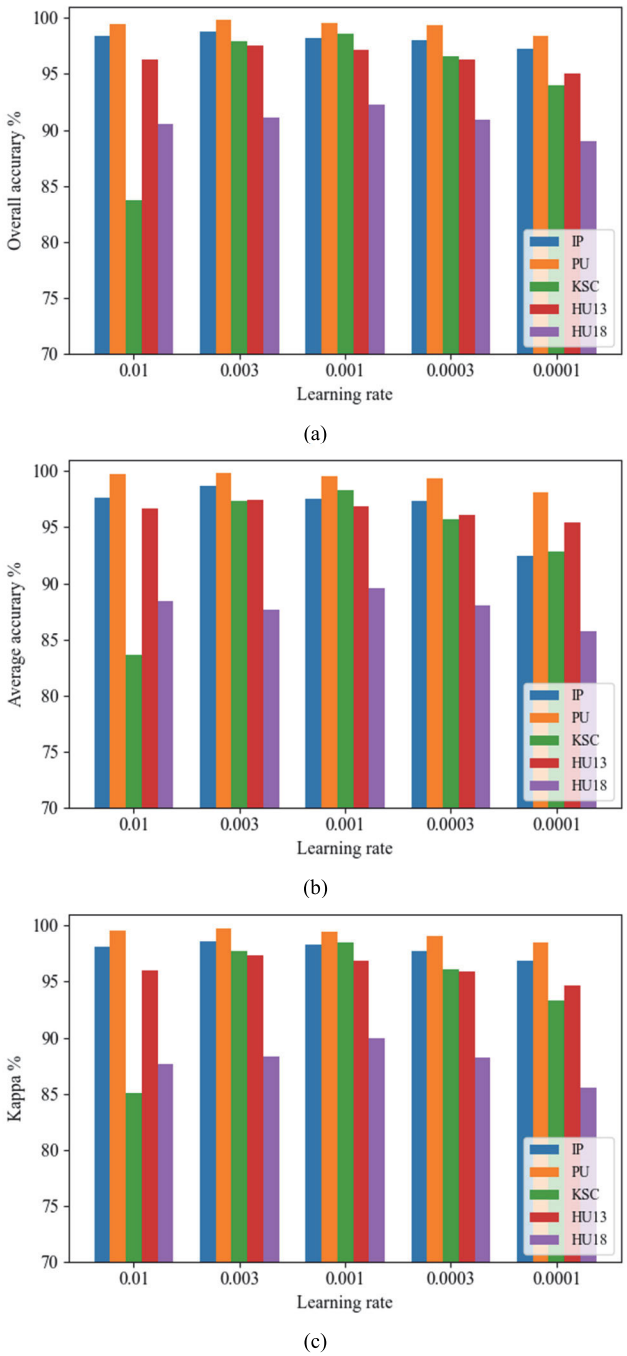


Fig. 10. (a)–(c) OA, AA, and Kappa coefficients of different learning rates, respectively.

potentially impact the model's performance. Hence, in this experiment, the optimal spatial window size for the five datasets was selected from 9 × 9 to 21 × 21, and the results are shown in Fig. 11(a)–(e). With an increase in the spatial window size, the classification accuracy of the IP dataset generally exhibits an upward trend but starts to decline slowly at a window size of 15 × 15. Although there is a subsequent increase, the larger window size would incur higher computational costs for the proposed model. Therefore, the spatial window size for the IP dataset was set to 15 × 15. Similarly, the spatial window size for the PU dataset also exhibits a gradual decrease after reaching 15 × 15, indicating that the input spatial window size for PU should be the same as that for IP. On the other hand, for the KSC dataset, the fluctuation trend becomes evident only after reaching 17 × 17, thus necessitating the setting of the input spatial window size for KSC as 17 × 17. For the HU13 dataset, the highest accuracy was noted at a spatial size of 11 × 11. However, as the spatial dimensions expanded, the accuracy gradually declined. Similarly, in the case of HU18, the peak precision was attained at a spatial size of 13 × 13, followed by a fluctuating downward trend.

D. Comparison of Different Algorithms

In order to ensure a fair evaluation of the performance and efficiency of the proposed model, it is compared with seven representative state-of-the-art deep learning models, namely, SSTN [45], SSAN [40], SSFTT [39], CVSSN [43], SSSAN [31], DSGSF [27], and MATA [28]. These selected algorithms cover a range of mainstream methods, including 2-D CNN structures, 3-D CNN structures, Transformer structures, spatial attention mechanisms, residual connections, dense connections, dual-branch architectures, and exploration of the central pixel contribution. To ensure a fair comparison, all algorithms are trained and tested on the same sample set, as shown in Tables I–V. Furthermore, both the proposed algorithm DBSSAN and the comparison algorithm adopt the experimental settings discussed above, including data preprocessing and hyperparameter settings, to ensure the comparability of the results.

1) Quantitative Analysis: To improve the accuracy of the comparison of classification results, the experimental results are averaged over ten runs. Tables VI–X provide the detailed information on the precision, OA, AA, and Kappa coefficients for each algorithm on the IP, KSC, PU, HU13, and HU18 datasets, along with their corresponding standard deviations. The optimal and suboptimal results are highlighted in boldface and underlined. The data clearly indicate that the proposed algorithm achieves the highest OA, AA, and Kappa coefficients, outperforming the other algorithms. On the IP dataset, the proposed DBSSAN demonstrates improvements of 0.11%, 0.43%, and 0.21% in the three evaluation indexes compared to the second-best algorithm DSGSF. Similarly, on the KSC dataset, the proposed algorithm surpasses the second-best algorithm by 0.24%, 0.20%, and 0.21% in terms of OA, AA, and Kappa. While the improvement on the PU dataset is not statistically significant, the proposed algorithm achieves the highest accuracy in all five categories. The proposed algorithm exhibits outstanding classification performance on the challenging HU13 and HU18 datasets. In particular, when compared to the second-best algorithm (CVSSN), the proposed algorithm demonstrates improvements of 0.51%, 0.42%, and 0.58% in OA, AA, and Kappa, respectively, on the HU13 dataset. On the HU18 dataset, the improvements are 0.91%, 0.76%, and 0.46%.

While machine learning predominantly relies on raw or shallow features for classification, deep learning has the capability to extract more abstract features. For instance, SSAN and SSSAN explore spatial features between the central pixel

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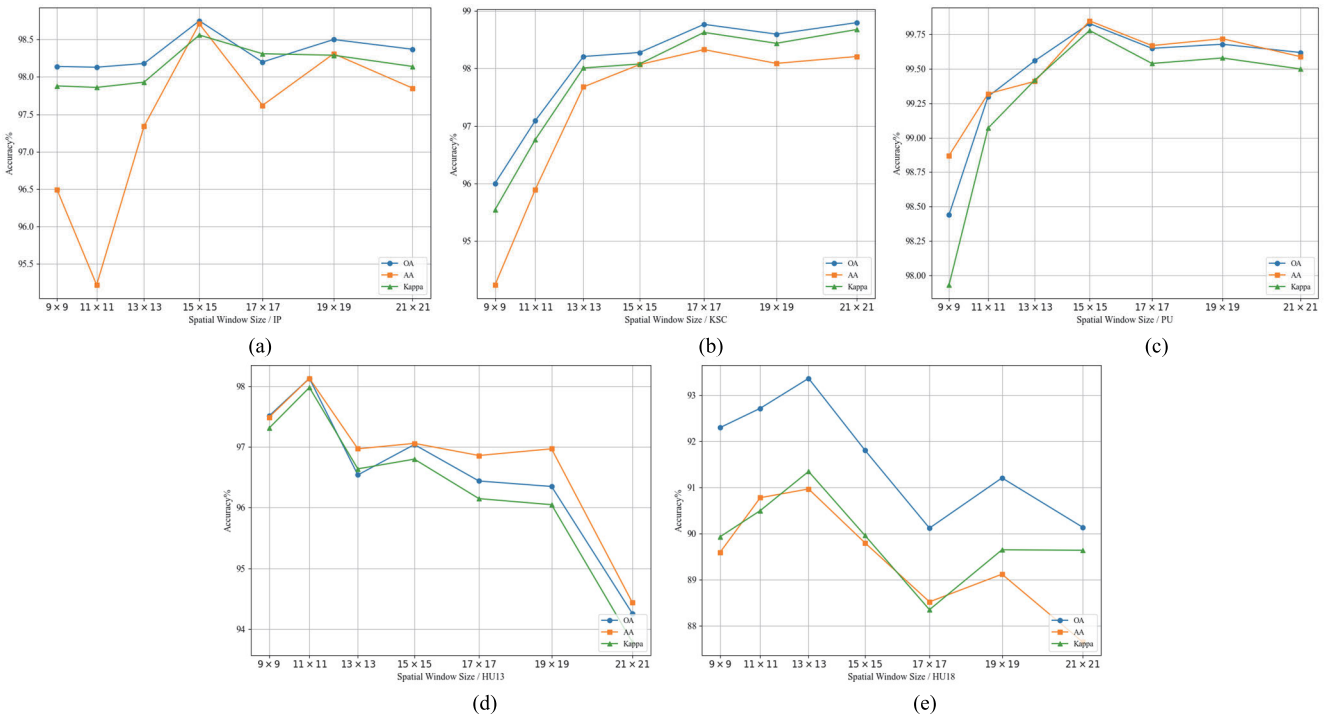


Fig. 11. (a)–(e) Changes of OA, AA, and Kappa coefficients with the input of spatial window sizes, respectively.

TABLE VI
CLASSIFICATION RESULTS OF DIFFERENT ALGORITHMS ON THE IP DATASET

Class	SSTN	SSAN	SSFTT	CVSSN	SSSAN	MATA	DSGSF	DBSSAN
1	92.5	**100**	86.96	90.91	85	**100**	**100**	**100**
2	91.98	90.77	97.78	96.66	97.34	97.74	**99.52**	97.73
3	91.54	92.22	**99.59**	98.25	98.92	99.32	97.53	99.06
4	**99.46**	97.88	91.32	98.58	94.01	98.54	98.99	96.52
5	95.39	94.61	98.54	98.6	98.16	98.96	**99.29**	97.95
6	98.29	92.46	98.75	99.52	99.84	**100**	**100**	**100**
7	75.86	81.82	52.17	92.31	**100**	84.24	94.11	**100**
8	98.39	98.16	100	99.77	98.14	**100**	**100**	**100**
9	72.73	61.54	93.33	88.89	**100**	**100**	**100**	**100**
10	98.8	95.98	98.58	98.82	99.29	95.31	97.6	**99.53**
11	93.28	93.92	97.58	98.36	98.58	98.24	98.55	**99**
12	91.45	91.03	96.18	95.7	**96.58**	98.54	96.41	94.09
13	**100**	98.37	**100**	**100**	96.81	99.12	**100**	**100**
14	98.48	99.26	**100**	98.25	99.29	99.85	99.73	**100**
15	98.78	90.64	98.54	99.42	98.85	99.05	95.85	**100**
16	93.1	**100**	97.56	96	97.53	**100**	96.38	92.41
OA (%)	95.09±0.59	95.32±0.94	97.99±0.15	98.14±0.16	98.42±0.16	98.55±0.02	98.64±0.21	**98.75±0.16**
AA (%)	91.78±1.41	95.68±1.19	94.38±0.32	96.40±0.62	97.29±0.59	97.57±0.04	97.58±0.17	**98.01±0.69**
K×100	94.40±0.63	95.32±1.09	97.71±0.17	97.88±0.18	98.31±0.19	98.28±0.02	98.45±0.13	**98.66±0.10**

and surrounding pixels through cosine similarity, CVSSN extracts features using both cosine and Euclidean similarity, and MATA utilizes a Transformer-based multihead attention mechanism to extract features from the central pixel and its adjacent pixel. Tables VI–VIII show that the proposed algorithms DBSSAN and CVSSN outperform SSAN and SSSAN for the KSC, PU, HU13, and HU18 datasets, indicating that the adoption of mixed similarity is superior to single similarity. Moreover, across multiple datasets, the proposed DBSSAN outperforms CVSSN, suggesting that the use of cosine and Gaussian–Euclidean similarities has a positive impact. Notably, for the IP and HU18 datasets, which have class imbalance, AA represents the average accuracy across all land cover categories. Improvements in AA for imbalanced datasets indicate an enhanced overall discriminative capability for each category, leading to more accurate identification and classification of minority group samples. It is worth highlighting that on the challenging and large-scale HU18 dataset, the proposed algorithm achieves a more significant improvement. The proposed algorithm's structure may be particularly well-suited for such challenging datasets. Overall, the proposed algorithm demonstrates a certain superiority across different datasets.

In addition, we also compare two newly proposed methods of cross-scale graph prototypical network (X-GPN) [46] and multiattention transformer (MAT) and adaptive

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TABLE VII
CLASSIFICATION RESULTS OF DIFFERENT ALGORITHMS ON THE KSC DATASET

Class	SSTN	SSAN	SSFTT	CVSSN	SSSAN	MATA	DSGSF	DBSSAN
1	98.83	96.43	**100**	**100**	**100**	98.97	**100**	**100**
2	94.44	83.47	99.54	88.43	99.09	96.32	97.57	**100**
3	99.55	97.1	94.02	99.13	**100**	**100**	**100**	96.1
4	82.01	80.38	94.14	**100**	94.37	93.39	92.37	99.1
5	86.21	90.62	95.56	96	85.63	98.99	**100**	99.3
6	89.59	**100**	96.21	**100**	**100**	95.48	95.95	**100**
7	94	**100**	98.95	**100**	**100**	98.99	**100**	**100**
8	95.76	95.36	99.74	**100**	99.22	92.16	98.2	97.14
9	95.03	94.25	99.78	**100**	**100**	99.48	99.56	99.36
10	98.85	95	99.72	98.6	**100**	99.78	99.71	**100**
11	98.68	**100**	**100**	**100**	**100**	98.59	98.15	**100**
12	95.52	92.16	99.33	96.14	99.56	**99.73**	98.65	99.33
13	**100**	99.88	**100**	**100**	**100**	100	100	**100**
OA (%)	95.93±0.24	93.09±1.58	98.82±0.16	98.75±0.05	98.36±0.46	98.54±0.03	98.83±0.11	**99.07±0.13**
AA (%)	94.05±0.44	91.39±2.05	97.98±0.26	98.34±0.04	97.51±0.61	97.97±0.07	98.47±0.14	**98.67±0.27**
K×100	95.47±0.27	92.30±1.76	98.66±0.20	98.61±0.06	98.18±0.51	97.92±0.05	98.71±0.09	**98.92±0.15**

TABLE VIII
CLASSIFICATION RESULTS OF DIFFERENT ALGORITHMS ON THE PU DATASET

Class	SSTN	SSAN	SSFTT	CVSSN	SSSAN	MATA	DSGSF	DBSSAN
1	99.27	97.21	99.84	99.87	**99.94**	97.99	98.99	99.78
2	99.86	99.05	99.46	**99.99**	99.71	99.58	99.93	99.93
3	92.34	98.43	99.78	98.94	99.8	97.08	99.94	**100**
4	97.07	99.82	97.74	98.86	98.55	99.55	**99.86**	99.72
5	99.27	**100**	99.3	99.61	98.99	99.76	**100**	**100**
6	**100**	97.32	97.08	99.92	97.17	96.72	**100**	99.56
7	99.68	97.84	99.84	**100**	**100**	98.28	**100**	**100**
8	95.62	95.24	98.09	98.64	98.86	97.89	97.53	**99.66**
9	91.41	98.99	94.92	99.33	99.89	99.89	99.88	**100**
OA (%)	98.62±0.15	98.24±0.76	98.91±0.06	99.74±0.03	99.51±0.12	98.69±0.03	99.58±0.06	**99.83±0.04**
AA (%)	97.51±0.25	97.98±0.84	98.37±0.08	99.48±0.05	99.55±0.13	98.97±0.07	99.57±0.05	**99.75±0.03**
K×100	98.17±0.20	97.67±1.00	98.54±0.08	99.66±0.04	99.39±0.12	98.95±0.06	99.45±0.03	**99.78±0.06**

TABLE IX
CLASSIFICATION RESULTS OF DIFFERENT ALGORITHMS ON THE HU13 DATASET

Class	SSTN	SSAN	SSFTT	CVSSN	SSSAN	MATA	DSGSF	DBSSAN
1	94.98	99.64	**100**	99.01	97.95	98.67	99.18	96.03
2	97.54	98.76	92.44	99.11	99.02	**99.91**	99..28	99.11
3	99.52	99.66	95.34	99.84	**100**	**100**	**100**	**100**
4	98.49	98.22	86.81	98.03	97.48	98.84	98.92	**99.81**
5	99.73	99.82	98.66	**100**	98.84	**100**	99.91	**100**
6	98.57	**100**	88.54	96.48	92.83	99.64	96.86	96.6
7	96.26	94.11	96.88	97.38	97.55	98.5	96.91	**99.2**
8	93.25	97.34	97	98.35	98.75	95.02	93.87	**98.52**
9	95.83	95.22	90.46	96.13	96.25	95.26	95.26	**98.31**
10	96.87	97.23	98.25	96.43	95.97	97.84	95.99	**100**
11	97.15	96.86	99.26	98.47	98.91	96.16	98.35	**99.27**
12	93.44	92.8	94.53	96.13	97.39	96.91	92.63	**97.17**
13	97.71	95.56	99.49	98.56	99.04	96.68	89.91	**100**
14	**100**	98.15	96.21	**100**	99.22	98.45	98.68	99.22
15	99.83	98.33	98.57	**100**	100	99.15	98.98	**100**
OA (%)	96.47±0.44	97.16±0.21	96.33±0.61	98.13±0.05	97.91±0.09	97.92±0.12	97.08±0.03	**98.64±0.24**
AA (%)	97.08±0.32	97.36±0.19	96.38±0.51	98.26±0.04	97.92±0.04	98.07±0.14	97.14±0.23	**98.68±0.18**
K×100	96.15±0.49	96.94±0.24	96.04±0.66	97.97±0.07	97.76±0.11	97.75±0.09	96.84±0.03	**98.55±0.26**

superpixel segmentation-based active learning (MAT-ASSAL) [47]. X-GPN utilizes diverse-scale neighborhoods to construct adjacency matrices and employs a multibranch framework to explore spectral–spatial features through graph convolution. MAT-ASSA combines Transformer and adaptive superpixel segmentation, modeling remote contextual dependencies between spectral–spatial embeddings with Transformer's self-attention module, and applying adaptive superpixel segmentation to integrate features into learning for improved spatial constraints. Using the PU dataset as an exemplar, our approach exhibits superior performance, notably achieving the maximal enhancement in terms of AA at 99.75 ± 0.03%. This serves to further validate the efficacy of our method in addressing minority samples.

2) Qualitative Analysis: Figs. 12–16 present the classification maps obtained by different algorithms on the IP,

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TABLE X
CLASSIFICATION RESULTS OF DIFFERENT ALGORITHMS ON THE HU18 DATASET

Class	SSTN	SSAN	SSFTT	CVSSN	SSSAN	MATA	DSGSF	DBSSAN
1	76.63	83.1	82.63	93.23	**95.84**	84.4	92.36	90.73
2	87.12	90.61	90.73	**95.86**	93.95	93.72	90.81	95.63
3	**100**	94.08	99.59	93.82	99.4	98.96	**100**	**100**
4	83.67	93.09	93.59	**95.99**	95	94.19	92.88	94.66
5	80.49	90.37	84.64	86.89	68.17	90.34	70.75	**90.86**
6	98.04	97.55	98.31	97.41	**99.15**	98.83	77.2	97.83
7	97.98	**100**	92	98.45	99.4	96.96	94.74	**100**
8	91.74	95.2	93.37	95.42	93.97	95.93	95.63	**96.64**
9	93.8	94.64	95.97	96.07	95.2	95.58	96.48	**96.64**
10	70.66	82.12	81.47	87.03	88.05	84.14	80.84	**90.6**
11	59.43	79.19	69.87	76.14	74.23	**81.09**	69.93	78.84
12	6.15	31.57	32.15	22.6	54.43	24.11	47.57	26.65
13	77.27	88.03	84.51	**89.02**	84.82	84.75	84.48	84.99
14	87.76	93.95	96.93	92.38	89.03	**97.03**	88.74	91.8
15	90.67	94.72	97.2	**98.75**	98.62	97.12	94.64	98.64
16	78.23	94.24	93.51	93.38	95.24	90.69	89.81	**95.91**
17	**100**	**100**	85.19	94.64	68.66	84.78	57.21	98.65
18	81.34	**94.71**	91.1	94.08	88.43	90.75	84.41	92.17
19	90.47	97.49	**98.96**	97.56	96.18	95.07	96.05	98.58
20	98.53	99.74	**99.65**	99.5	99.03	99.29	97.54	99.58
OA (%)	89.59±0.03	91.43±0.08	90.79±0.15	92.03±0.19	91.65±0.06	92.02±0.03	90.64±0.06	**92.94±0.29**
AA (%)	83.42±0.58	89.68±0.05	87.53±0.06	89.80±0.12	89.40±0.56	88.89±0.02	85.82±0.75	**90.56±0.36**
K×100	82.49±0.02	88.77±0.11	88.04±0.24	90.24±0.39	89.26±0.24	89.57±0.04	88.27±0.38	**90.70±0.45**

KSC, PU, HU13, and HU18 datasets, along with false-color composite images and real terrain maps for direct comparison. Although SSTN, SSAN, SSFTT, CVSSN, SSSAN, MATA, DSGSF, and the proposed algorithm DBSSAN produce similar classification maps, DBSSAN achieves the clearest and most visually similar classification map to the real world. In the "Gravel" class of the PU dataset, the classification closely resembles the ground truth land cover map, with an accuracy of 100%, as indicated in Table VIII. Moreover, on the HU13 and HU18 datasets, the identification of the circular runway closely approximates the actual land cover map, while other algorithms tend to misclassify pixels in this region to varying extents. Comparatively, DBSSN produces the most accurate and smooth classification maps, particularly along the boundaries of different classes. The IP dataset, characterized by dense land cover, presents a more challenging classification task. Upon observing the lower left portion of the classification map generated by this algorithm, it becomes evident that the classification of class boundaries is the most accurate. The IP dataset presents a more challenging classification task due to its dense land cover. Notably, upon observing the lower left portion of the classification map generated by this algorithm, it becomes evident that the classification of class boundaries is the most accurate.

3) Efficiency Analysis: To compare the different methods in terms of model efficiency, Table XI records the running time, model complexity of the different algorithms, and computational cost. Table XI shows that the training time of DBSSAN is marginally slower than that of SSFTT and CVSSN. This can be attributed to the fact that SSFTT and CVSSN both utilize a single-branch structure, while DBSSN employs a dual-branch structure, resulting in a relatively slower training process. However, the model complexity of DBSSN falls between these two, which is a commendable outcome. Other methods may have longer training times due to the adoption of a dual-branch structure or 3-D CNNs with a substantial number of parameters. For example, SSAN employs numerous 3-D CNNs, making it the most complex model. DSGSF follows a dual-branch structure, and its spectral features are extracted by an LSTM model. Due to the inherent structure of the LSTM model, it incurs a relatively higher training cost. Although SSTN has the minimum model complexity, its precision is not notably high, as indicated by the experimental results. In contrast, despite the slightly higher model complexity of DBSSAN compared to SSTN, considering its acceptable performance in terms of accuracy, this minor sacrifice is deemed worthwhile. Concurrently, it is evident that DBSSAN's FLOPs rank second across all datasets, indicating a lower computational cost for this algorithm. This suggests that, with relatively modest computational overhead, DBSSAN achieves commendable performance, presenting an efficient and feasible solution for HSI classification tasks. This balanced design, which takes into account both computational efficiency and performance, renders DBSSAN particularly attractive for practical applications, particularly in scenarios characterized by limited resources or high demands for computational efficiency.

Overall, the proposed algorithm DBSSAN may not be the optimal choice in terms of runtime, model complexity, and computational cost. However, its outstanding high accuracy compensates for this drawback.

E. Comparison With Popular Attention Mechanisms

To evaluate the impact of the proposed CG-SA on classification performance, prevalent attention mechanisms were employed as substitutes for CG-SA on imbalanced IP datasets and the demanding HU13 dataset. These alternative attention mechanisms encompassed the channel attention mechanism (SE) [33], spatial attention mechanism in CBAM (SAM) [34], spatial group enhancement module (SGEM) [48], and global attention mechanism (GAM) [49]. The experimental

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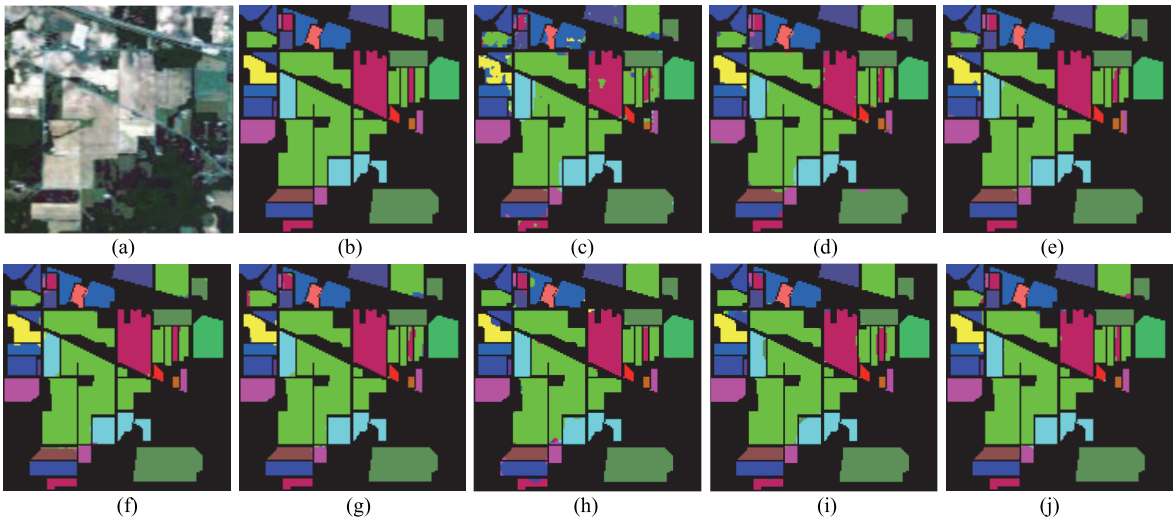


Fig. 12. Classification maps of IP dataset. (a) False-color map. (b) Ground truth map. (c) SSTN. (d) SSAN. (e) SSFTT. (f) CVSSN. (g) SSSAN. (h) MATA. (i) DSGSF. (j) DBSSAN.

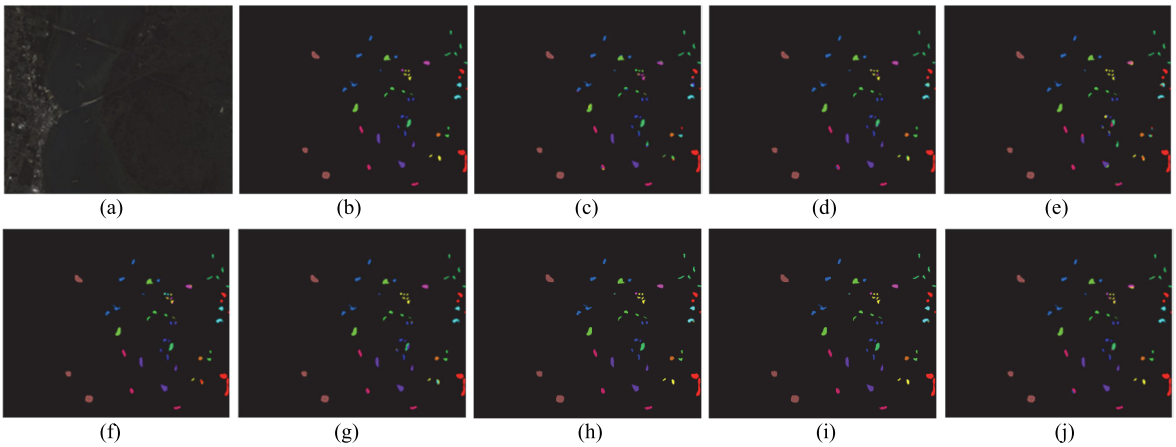


Fig. 13. Classification maps of KSC dataset. (a) False-color map. (b) Ground truth map. (c) SSTN. (d) SSAN. (e) SSFTT. (f) CVSSN. (g) SSSAN. (h) MATA. (i) DSGSF. (j) DBSSAN.

TABLE XI

COMPARISONS ON MODEL EFFICIENCY OF DIFFERENT METHODS ON IP, KSC, PU, HU13, AND HU18 DATASETS

Dataset		SSTN	SSAN	SSFTT	CVSSN	SSSAN	MATA	DSGSF	DBSSAN
IP	Ttrain (m)	63.06	151.44	**28.42**	32.02	415.69	250.74	180.69	53.13
	Ttest (ms)	23.82	144.9	13.09	32.85	380.95	**5.41**	32.96	13.14
	Params (M)	**0.11**	23.47	0.61	1.02	1.02	2.28	0.18	0.34
	FLOPs (M)	**3.36**	605.80	16.91	58.48	26.72	27.68	100.34	10.87
KSC	Ttrain (m)	37.83	376.12	**17.46**	22.69	257.62	178.29	144.60	34.18
	Ttest (ms)	27.94	489.47	22.49	57.52	409.14	**6.15**	162.54	24.89
	Params (M)	**0.10**	58.79	0.42	1.00	1.09	2.20	0.24	0.31
	FLOPs(M)	**4.37**	347.19	13.49	74.19	29.44	28.79	1701.14	11.12
PU	Ttrain (m)	130.27	90.28	**55.24**	61.32	965.2	507.08	462.15	109.71
	Ttest (ms)	23.16	33.48	9.37	21.44	319.39	**5.39**	96.46	11.25
	Params (M)	**0.09**	14.71	0.16	0.97	0.98	2.09	0.18	0.28
	FLOPs (M)	**3.08**	46.63	13.21	69.24	19.83	21.07	864.68	6.81
HU13	Ttrain (m)	89.29	61.64	**41.89**	50.13	640.75	164.77	595.52	83.86
	Ttest (ms)	16.23	16.93	8.2	18.66	350.82	**4.32**	305.47	10.86
	Params (M)	**0.07**	8.21	0.15	0.98	0.98	2.12	0.19	0.28
	FLOPs (M)	**1.50**	25.08	5.95	30.57	14.72	14.96	2800.42	5.27
HU18	Ttrain (m)	639.02	424.11	**295.49**	525.9	5090.63	1685.43	6954.83	557.1
	Ttest (ms)	20.21	20.11	8.6	12.02	131.42	**5.33**	768.14	16.93
	Params (M)	**0.07**	11.3	0.18	0.98	0.98	2.16	0.19	0.26
	FLOPs (M)	**2.21**	42.53	8.32	37.65	13.77	18.32	6096.94	4.07

findings, as delineated in Table XII, reveal a notable enhancement in classification performance with CG-SA compared to these alternative attention mechanisms. Specifically, on the IP dataset, CG-SA exhibits a marked improvement for class

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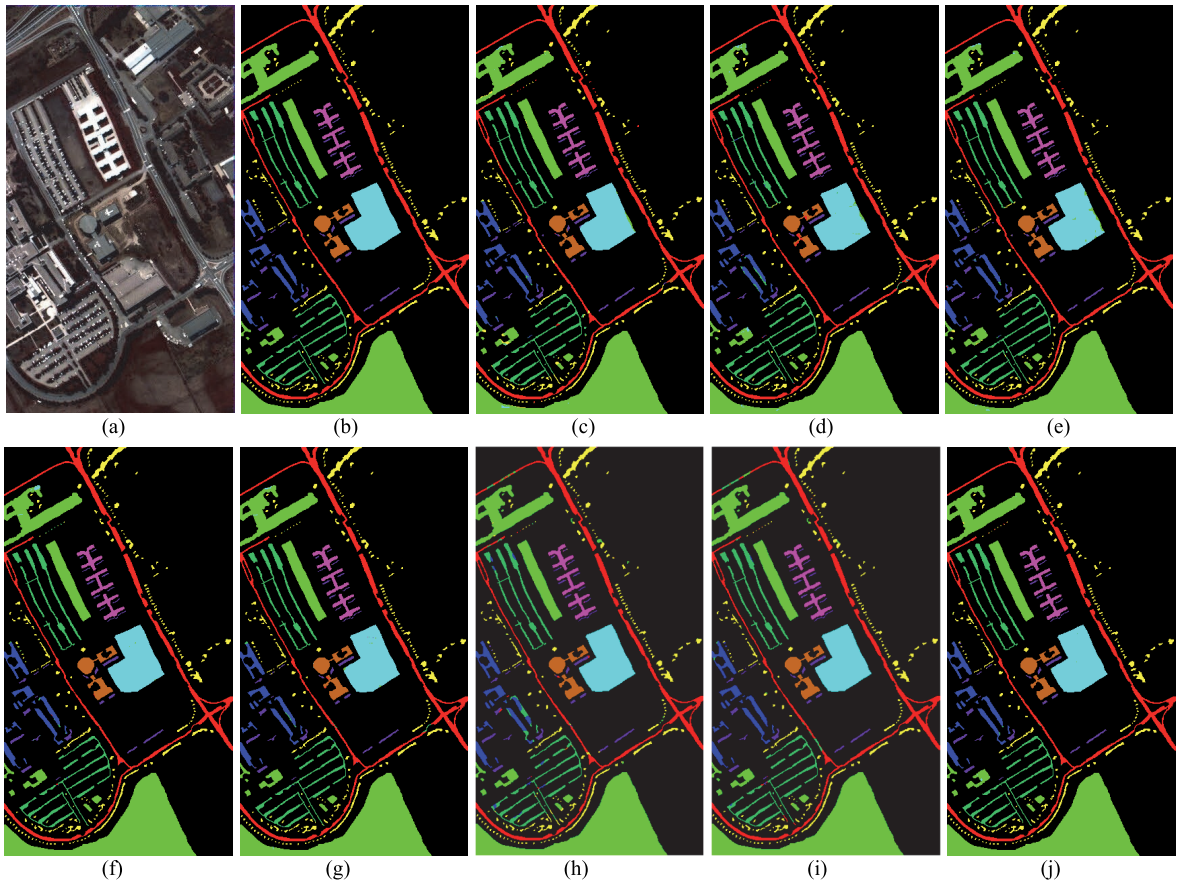


Fig. 14. Classification maps of PU dataset. (a) False-color map. (b) Ground truth map. (c) SSTN. (d) SSAN. (e) SSFTT. (f) CVSSN. (g) SSSAN. (h) MATA. (i) DSGSF. (j) DBSSAN.

TABLE XII
CLASSIFICATION EFFECT OF DIFFERENT ATTENTION MECHANISMS ON THE CLASSIFICATION OF IP AND HU13 DATASETS

Dataset		OA (%)	AA (%)	K×100
IP	CG-SA	**98.75±0.16**	**98.01±0.69**	**98.66±0.10**
	SE	98.58±0.07	96.77±0.40	98.41±0.10
	CBAM	98.60±0.12	96.65±0.35	97.82±0.14
	SGEM	98.36±0.21	97.88±0.78	97.86±0.17
	GAM	97.55±0.11	96.06±0.13	97.21±0.12
HU13	CG-SA	**98.64±0.24**	**98.68±0.18**	**98.55±0.26**
	SE	97.60±0.19	97.64±0.14	97.40±0.22
	CBAM	97.76±0.53	97.78±0.38	97.52±0.65
	SGEM	98.26±0.12	98.19±0.28	98.22±0.14
	GAM	97.92±0.16	97.81±0.25	97.76±0.17

A, indicating its adeptness in handling minority class samples and thereby enhancing overall classification performance. On the more challenging HU13 dataset, CG-SA surpasses other attention mechanisms significantly across all three evaluation metrics, underscoring that CG-SA, by delving into the feature relationships between central and surrounding pixels, more effectively extracts features, culminating in superior performance in classification tasks. These findings underscore that CG-SA offers an effective and viable solution for HSI classification tasks.

F. Ablation Study

In order to thoroughly investigate and validate the effectiveness of the proposed DBSSAN model, this section primarily discusses the impact of Gaussian–Euclidean distance similarity and each module on the classification performance. The experiments were conducted under the same conditions for ten runs, and the results show the average and standard deviation. The best results are highlighted in bold.

1) Impact of Gaussian–Euclidean Distance Similarity: In order to examine the influence of the proposed Gaussian–Euclidean similarity on the relationship between the central pixel and its surrounding pixels, we utilized the same algorithmic framework to compare single cosine similarity (Cos), cosine and Euclidean similarity (Cos + ED), and the cosine and Gaussian–Euclidean similarity (Cos + GaED) proposed here. According to Table XII, it can be deduced that the use of mixed similarity significantly outperforms single similarity metrics, particularly the Cos + GaED utilized in HU13 and HU18 datasets, demonstrating a notable accuracy improvement of approximately 1%. This suggests that the use of mixed similarity effectively mitigates the insensitivity of cosine similarity to vector length. When comparing Cos + GaED with Cos + ED, it is evident that the utilization of Gaussian–Euclidean similarity outperforms Euclidean similarity. Therefore, the Gaussian kernel function has a positive impact on handling complex, multidimensional HSIs, allowing for a more comprehensive extraction of the relationship between central and surrounding pixels, thereby enhancing feature extraction. It is important to note that the IP and HU18 datasets exhibit class imbalance, where AA represents the average accuracy across all land cover categories. For

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TABLE XIII
INFLUENCE OF DIFFERENT SIMILARITY ON THE RELATION BETWEEN CENTRAL PIXEL AND SURROUNDING PIXELS

Spatial attention mechanism		IP	KSC	PU	HU13	HU18
Cos	OA (%)	98.56±0.05	98.8±0.05	99.58±0.01	97.6±0.23	92.04±0.82
	AA (%)	96.75±0.48	98.26±0.06	99.63±0.03	97.59±0.29	89.54±0.68
	K×100	98.36±0.06	98.67±0.05	99.44±0.01	97.4±0.25	89.62±0.65
Cos+ED	OA (%)	98.37±0.12	98.92±0.06	99.65±0.06	97.78±0.23	92.17±0.20
	AA (%)	97.05±0.18	98.61±0.16	99.67±0.08	97.92±0.22	89.45±0.66
	K×100	98.28±0.13	98.78±0.13	99.57±0.08	97.61±0.43	89.82±0.33
Cos+GaED	OA (%)	**98.75±0.16**	**99.07±0.13**	**99.83±0.04**	**98.64±0.24**	**92.94±0.29**
	AA (%)	**98.01±0.69**	**98.67±0.27**	**99.75±0.03**	**98.68±0.18**	**90.56±0.36**
	K×100	**98.66±0.10**	**98.92±0.15**	**99.78±0.06**	**98.55±0.26**	**90.70±0.45**

TABLE XIV
IMPACT OF EACH MODULE ON CLASSIFICATION PERFORMANCE

Dataset	CG-SA	SIE	Transformer	OA (%)	AA (%)	K×100
IP	√	√	√	**98.75±0.16**	**98.01±0.69**	**98.66±0.10**
	×	√	√	98.49±0.11	95.79±0.71	98.28±0.12
	√	×	√	98.35±0.15	97.03±0.32	98.11±0.17
	√	√	×	98.21±0.05	95.46±0.28	97.95±0.07
KSC	√	√	√	**99.07±0.13**	**98.67±0.27**	**98.92±0.15**
	×	√	√	98.99±0.06	98.27±0.17	98.88±0.06
	√	×	√	98.71±0.20	98.57±0.51	98.68±0.27
	√	√	×	98.84±0.07	98.05±0.13	98.71±0.08
PU	√	√	√	**99.83±0.04**	**99.75±0.03**	**99.78±0.06**
	×	√	√	99.52±0.09	99.49±0.10	99.38±0.13
	√	×	√	98.42±0.12	99.13±0.14	99.44±0.07
	√	√	×	99.24±0.03	99.46±0.07	98.99±0.04
HU13	√	√	√	**98.64±0.24**	**98.68±0.18**	**98.55±0.26**
	×	√	√	96.27±0.52	96.52±0.27	95.99±0.65
	√	×	√	97.62±0.25	97.89±0.23	97.41±0.29
	√	√	×	97.36±0.37	97.55±0.31	97.15±0.40
HU18	√	√	√	**92.94±0.29**	**90.56±0.36**	**90.70±0.45**
	×	√	√	91.84±0.43	88.29±0.32	89.42±0.21
	√	×	√	89.40±0.14	89.10±0.87	87.52±1.13
	√	√	×	91.58±0.56	86.34±0.33	89.19±0.64

imbalanced datasets, the increase in AA signifies an overall improvement in discriminative ability for each category. Thus, this algorithm demonstrates a more accurate identification and classification of minority group samples.

In summary, the proposed cosine and Gaussian–Euclidean similarity in this article appears to better handle the complexity of hyperspectral data and the relationships between features. This results in improved accuracy and stability in classification tasks.

2) Impact of Each Module: As shown in Fig. 14, it is evident that the absence of CG-SA leads to a significant decrease in AA on the IP, HU13, and HU18 datasets, with reductions of 2.22%, 2.16%, and 2.17%, respectively. As mentioned earlier, AA represents the overall discriminative ability of the model for each category, highlighting that the contribution of CG-SA mainly lies in enhancing the model's feature extraction capabilities for each category. When the SIE module is omitted, the model's performance declines across all datasets. This decrease can be attributed to CG-SA's effectiveness in extracting global features but having a weaker ability to capture local details. However, the incorporation of the SIE significantly enhances the model's performance. This improvement can be attributed to the advantages of global–local fusion. While CG-SA models global features through nonlocal operations, the SIE module extracts multiscale local information, enabling the model to comprehensively consider both global and local features. The integration of both global and local information results in a noteworthy improvement in the classification accuracy of the model, particularly on the HU18 dataset, with respective enhancements of 3.54%, 1.46%, and 3.18% for the OA, AA, and Kappa coefficients, respectively. The charts also clearly reveal that the absence of the Transformer module leads to a substantial decrease in AA on the IP and HU18 datasets. This underscores the crucial importance of extracting spectral features for model performance. Spectra contain a wealth of crucial information, emphasizing the need to thoroughly extract these spectral features and combine them with spatial features to enhance the overall discriminative ability of the model. Concurrently, we explored the influence of incorporating SIE convolutional kernels. For example, $(1 + 3 + 5 + 7)$ denotes the inclusion of 5×5 and 7×7 convolutional kernels in addition to the existing 1×1 and 3×3 kernels. The experimental results are depicted in Fig. 17. It is evident that the addition of convolutional kernels does not result in an enhancement in AA and may even impact the accuracy of AA. Moreover, the augmentation of SIE kernels also amplifies the model's complexity.

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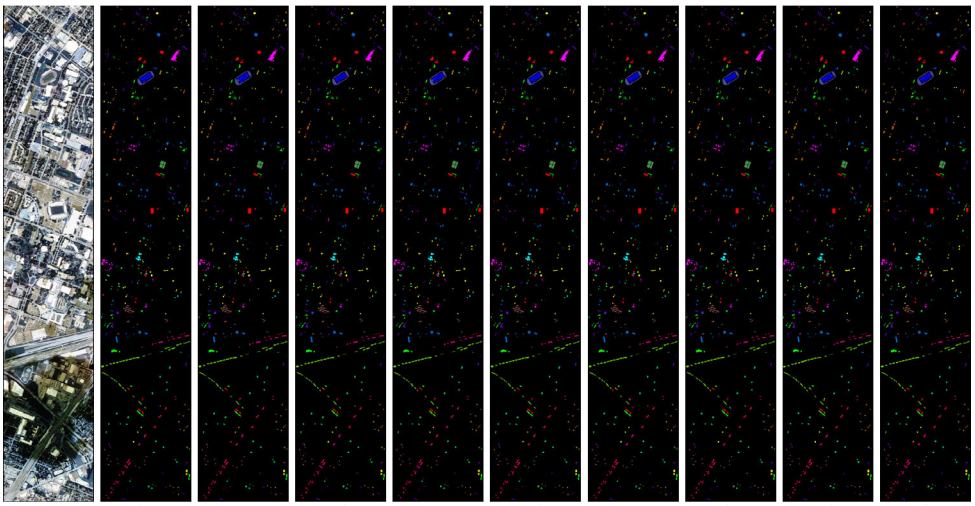


Fig. 15. Classification maps of HU13 dataset. (a) False-color map. (b) Ground truth map. (c) SSTN. (d) SSAN. (e) SSFTT. (f) CVSSN. (g) SSSAN. (h) MATA. (i) DSGSF. (j) DBSSAN.

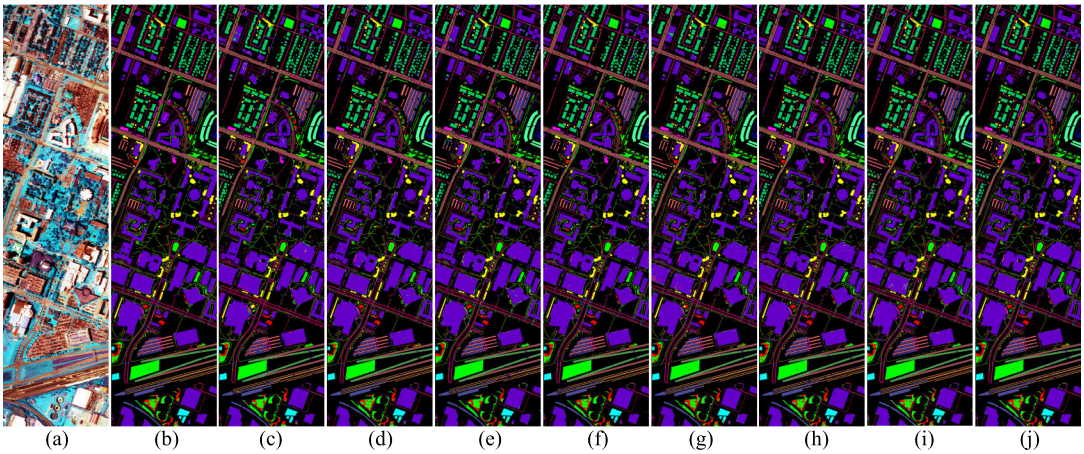


Fig. 16. Classification maps of HU18 dataset. (a) False-color map. (b) Ground truth map. (c) SSTN. (d) SSAN. (e) SSFTT. (f) CVSSN. (g) SSSAN. (h) MATA. (i) DSGSF. (j) DBSSAN.

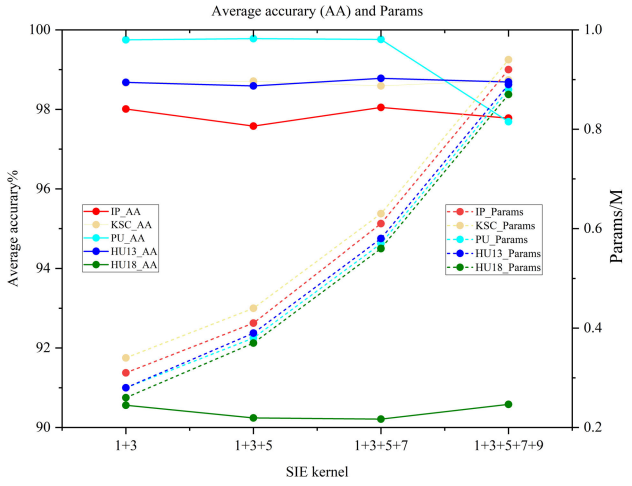


Fig. 17. Effect of increasing SIE kernel on AA and Params.

IV. Conclusion

This article presents a dual-branch strategy aimed at reducing computational complexity and costs while maintaining computational efficiency. The incorporation of a novel spatial attention mechanism, which explores the similarity between central pixels and their surrounding counterparts, has effectively improved the accuracy of central pixel recognition and mitigated dimensionality issues associated with hyperspectral imaging (HSI). In addition, this approach integrates both global and local features to enhance feature representation and classification performance. By employing a Transformer to extract spectral information concurrently, the model captures the distance dependence of hyperspectral data and nonlinear correlations, thereby enhancing its understanding of spectral characteristics. The experimental results and analysis demonstrate that the proposed DBSSAN model exhibits commendable performance and efficiency.

However, despite achieving satisfactory results to some extent, there are still limitations to the method presented in this article. Moreover, the complex network architecture necessitates a larger number of training samples, which may pose challenges in practical applications. Therefore, our future research will focus on further enhancing the network to achieve outstanding classification performance even when faced with a limited number of samples.

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